Product Strategy & Roadmap: AI Search Visibility / Generative Engine Optimization (GEO)

Prepared by: Product Management Pod, ProManage
Resource Constraints: 2 Engineers | 1 Quarter (12 Weeks Total | 4 Weeks per Phase)

1. Executive Summary & Strategic Context

As search behavior shifts from keyword-based listings to structured AI answers (ChatGPT, Gemini, Perplexity, Google Search Generative Experiences), enterprise multi-location brands like Raymond, Muthoot, and EuroKids face an acute threat: **AI invisibility or hallucinated business info**. When consumers ask AI engines for regional touchpoints, operating hours, or specific services, accurate retrieval directly impacts trust and brick-and-mortar footfall.

ProManage's Unique Core Advantage: We already hold the underlying structured golden records (verified location nodes, structural local schemas, operational variables, and review feeds) for thousands of retail touchpoints. This roadmap outlines how to transform this static data moat into a highly defensible, high-margin, iterative AI Search Visibility suite inside 12 weeks using only 2 dedicated engineers.

2. Core Product Architecture & Scoping Framework

Given the resource ceiling (2 engineers), we strictly avoid heavy infrastructure or complex automated data sync pipelines on Day 1. Instead, the roadmap focuses on a pragmatic progression: **Audit & Diagnostic -> Actionable Synchronization -> AI Performance Intelligence**.

Phase / Window	Core Theme & Focus	Engineering Scoping Strategy
MVP (Weeks 1-4)	AI Brand Share & Baseline Diagnostic Audit	Manual parsing scripts, fixed location samples, hardcoded benchmarking templates.
V1 (Weeks 5-8)	Active Schema Injection & LLM Optimisation	Dynamic API payload expansion, real-time structured JSON-LD engine, local crawler optimizations.
V2 (Weeks 9-12)	GEO Performance Dashboards & Insights Engine	Synthetic query monitoring automation, citation tracking, high-value keyword alerting.

3. Phase 1: The MVP (Weeks 1–4) — AI Brand Share & Diagnostic Audit

Objective: Build a light diagnostic utility that demonstrates AI citation risk to enterprise stakeholders, generating an immediate commercial hook.

Core Capabilities & Functional Requirements:

- **Synthetic Search Engine:** A backend runner utilizing standard text-generation APIs (OpenAI/Gemini) to programmatically ask localized queries (e.g., 'Where is the nearest Raymond store for formal suits in South Mumbai?').
- **Citation Matching Engine:** Simple regex and keyword string matching to determine if the customer's brand name, exact address, or telephone line appears in the engine's structured answer/citations.
- **PDF Executive Explainer:** A basic, template-driven report generator highlighting an 'AI Visibility Score' (Percentage of accurate citations captured vs missed across a sample group of 20 high-priority locations).

Engineering Architecture & Pragmatic Execution:

To ensure shipping in 4 weeks with 2 engineers, we eliminate background message queues (like Celery) and real-time frontend charts. The backend will execute queries synchronously or via basic bash cron loops. The user interface will be simple: an account executive uploads a CSV containing 20 core store nodes, and the system runs a batch script, outputting a static download link for the PDF diagnostic report.

Success Metrics & Milestones:

- **Week 2 Milestone:** Core synthetic prompt runner working via postman/terminal for ChatGPT and Gemini.
- **Week 4 Milestone:** End-to-end PDF report generation functioning smoothly for internal sales demos.
- **North Star Metric:** Technical stability of prompt execution across a 20-location batch run under 180 seconds.

4. Phase 2: V1 (Weeks 5–8) — Schema Optimization & LLM Injection Engine

Objective: Move from passive scanning to active optimization. Build features that dynamically maximize the structure, authority, and discovery readiness of the brand's local web assets.

Core Capabilities & Functional Requirements:

- **GEO-Optimised JSON-LD Engine:** Automatically embed localized structured markup containing precise attributes (GeoCoordinates, openingHours, specific services offered, reviews, and entity relationships) onto the client's local store locator landing pages.
- **AI Context Layer:** A clean, un-styled, machine-readable endpoint (e.g., /ai-fact-sheet) for each store location node, explicitly optimized for rapid indexing by Perplexity, OpenAI, and Google crawlers.
- **Real-time Business Attribute Verification:** A UI matrix within ProManage allowing branch managers to override or flag discrepancies in baseline operational data immediately before it gets crawled.

Engineering Architecture & Pragmatic Execution:

Instead of building real-time synchronization pipelines with third-party AI APIs (which are often closed or slow), the engineers will focus fully on managing the assets ProManage controls directly. We will output an optimized, cacheable JSON file per brand location node that clients can inject instantly via their existing Google Tag Manager (GTM) setups or basic reverse-proxy rules. This avoids engineering dependencies on the client's core IT infrastructure.

Success Metrics & Milestones:

- **Week 6 Milestone:** Automated generation of machine-readable schema layers validated via schema markup testing suites.
- **Week 8 Milestone:** Live GTM delivery script tested and active for a pilot customer group.
- **North Star Metric:** 100% compliance of injected schemas against W3C and major search engine crawling syntax specifications.

5. Phase 3: V2 (Weeks 9–12) — Performance Intelligence & GEO Dashboard

Objective: Provide comprehensive, enterprise-grade, continuous visibility tracking, shifting the product from an engineering project to a sticky SaaS platform module.

Core Capabilities & Functional Requirements:

- **Multi-Engine Tracking Dashboard:** A unified web UI chart displaying trends in visibility share across ChatGPT, Gemini, Perplexity, and Google AI over time.
- **Sentiment & Competitor Context Parsing:** Advanced text extraction from AI engine responses to isolate competitor mentions (e.g., if Muthoot is mentioned, identify if Manappuram is recommended alongside them).
- **Critical Misinformation Alert System:** An automated alerting framework that triggers internal email notices if an AI engine hallucinates a 'permanently closed' status or displays an incorrect emergency helpline.

Engineering Architecture & Pragmatic Execution:

To shield our 2 engineers from complex database maintenance, we will avoid large elasticsearch setups. The dashboard analytics will be powered by daily background cron aggregates stored directly in a relational table (PostgreSQL jsonb column). Charts will be drawn using light, open-source charting components (e.g., Chart.js) embedded right into the existing ProManage merchant platform.

Success Metrics & Milestones:

- **Week 10 Milestone:** Database cron jobs and jsonb storage structures finalized and operational.
- **Week 12 Milestone:** Interactive dashboard UI fully integrated and launched to production.
- **North Star Metric:** Customer retention/engagement (at least 60% of pilot accounts log in weekly to check GEO volatility).

6. Operational Playbook & Risk Management

Handling Rate Limits & Network Blocks: AI platforms frequently alter search rules and scrape limits. The engine must randomize search execution times and rotate basic proxy wrappers to protect diagnostic reliability.

Dealing with Unclean Client Data: Enterprise databases are notoriously dirty. The system will run a mandatory data pre-validation filter during the MVP phase. If a location lacks an explicit postal index number or geocode, it is automatically held in a staging queue rather than running broken diagnostic queries.